

[illegible]

1. Tiotropium donkey (LAMA, first-line)

WHAT YOU SEE

Donkey labeled 'TIOTROPIUM' with trophy, '↓exacerbations', '↓symptoms', 'FEV1 decline halted'.

WHAT IT ENCODES

Tiotropium is a long-acting muscarinic antagonist (LAMA) and a foundational COPD maintenance therapy: it blocks M3 receptors → bronchodilation, reduces exacerbations and symptoms, and improves quality of life. LAMA/LABA long-acting bronchodilators are the backbone of treatment across groups.

BOARD TRAP

Maintenance bronchodilators relieve symptoms and reduce exacerbations but do NOT reverse fixed airflow obstruction or reduce mortality. Only smoking cessation and long-term oxygen (in hypoxemia) have proven mortality benefit.

2. No evidence dust mask

WHAT YOU SEE

Dust mask labeled 'NO EVIDENCE — wrong answer'.

WHAT IT ENCODES

Generic mucolytics and dust masks are not proven disease-modifying therapy in COPD and are common distractor 'wrong answers.' Effective interventions are smoking cessation, vaccination, bronchodilators, pulmonary rehab, and LTOT when indicated.

BOARD TRAP

Choosing mucolytics or a 'dust mask' as a core COPD therapy is the wrong board answer.

3. Group A — bronchodilator alone

WHAT YOU SEE

GROUP A panel showing a single long-acting bronchodilator as first-line.

WHAT IT ENCODES

GOLD Group A (low symptoms: mMRC 0-1 / CAT <10, AND 0-1 moderate exacerbations with no hospitalization) starts with a SINGLE long-acting bronchodilator (LAMA or LABA). The ABE grouping — driven by symptom burden and exacerbation history — selects initial inhaler therapy, not the spirometric grade.

BOARD TRAP

Initial inhaler choice is by ABE group (symptoms + exacerbations), not by GOLD 1-4 FEV1 grade.

4. Group B — LAMA+LABA

WHAT YOU SEE

GROUP B panel with dual bronchodilator (LABA + LAMA) inhaler robots.

WHAT IT ENCODES

GOLD Group B (more symptoms: mMRC ≥2 / CAT ≥10, but still low exacerbation risk) is treated with DUAL bronchodilation: LABA + LAMA combination, which outperforms monotherapy on dyspnea and lung function.

BOARD TRAP

Do NOT add an inhaled corticosteroid first in Group B — dual LABA+LAMA bronchodilation is preferred; ICS is reserved for exacerbators/high eosinophils.

5. Group E — LAMA+LABA, add ICS if eos ≥300

WHAT YOU SEE

GROUP E panel: 'LAMA + LABA (+/- ICS)', 'ADD ICS if eosinophils ≥300'.

WHAT IT ENCODES

GOLD Group E (frequent/severe Exacerbations: ≥2 moderate or ≥1 hospitalization in a year) starts with LABA + LAMA; ADD an inhaled corticosteroid (triple therapy) when blood eosinophils are ≥ 300 cells/μL. ICS strongly favored if eos ≥300 or concomitant asthma.

BOARD TRAP

Add ICS based on high eosinophils and exacerbation history — NOT as routine therapy and never as monotherapy. ICS increases PNEUMONIA risk, so it is targeted, not universal.

6. ICS never alone

WHAT YOU SEE

Inhaler robots with bold 'NEVER ALONE' sign for inhaled corticosteroid.

WHAT IT ENCODES

Inhaled corticosteroids in COPD are ALWAYS paired with a long-acting bronchodilator (LABA+ICS or triple LABA+LAMA+ICS) — never used as monotherapy, unlike in asthma. The benefit of ICS is reducing exacerbations in the right phenotype.

BOARD TRAP

ICS monotherapy is NEVER appropriate in COPD. Also weigh the pneumonia risk of ICS against exacerbation reduction.

7. Eosinophil counting station

WHAT YOU SEE

Counting station tracking blood eosinophils as the threshold for adding ICS.

WHAT IT ENCODES

Blood eosinophil count is a biomarker guiding ICS use: benefit is greatest at ≥ 300 cells/μL, intermediate 100-300, and little benefit <100 cells/μL. It helps predict who will respond to inhaled steroids and who is at higher pneumonia risk for marginal gain.

BOARD TRAP

Low eosinophils (<100) predict POOR ICS benefit and favor de-escalation — do not add ICS reflexively in low-eosinophil patients.

8. Beta-blocker contraindication crossed

WHAT YOU SEE

Crossed-out warning panel: 'CONTRAINDICATION — less effective in active smokers'.

WHAT IT ENCODES

Active smoking blunts the anti-inflammatory response to inhaled corticosteroids. Smoking cessation is THE key disease-modifying step — it slows the accelerated FEV1 decline and is one of only two interventions (with LTOT in hypoxemia) shown to reduce mortality.

BOARD TRAP

ICS are less effective in current smokers; the highest-yield intervention is smoking cessation, not escalating inhalers. (Note: cardioselective beta-blockers are NOT contraindicated in COPD when otherwise indicated.)

9. Panlobular emphysema lungs (lower lobe)

WHAT YOU SEE

ALPHA-1-ANTITRYPSIN station with panlobular lower-lobe emphysema vs smoking-related upper-lobe.

WHAT IT ENCODES

Alpha-1 antitrypsin (AAT) deficiency causes PANACINAR (panlobular), LOWER-lobe predominant emphysema, classically in YOUNG (<45) non-smokers or those with disproportionate disease. AAT normally inhibits neutrophil elastase; its deficiency lets elastase destroy alveolar walls. Screen with a serum AAT level.

BOARD TRAP

Lower-lobe panacinar emphysema in a young/non-smoking patient = alpha-1, NOT typical smoking COPD (which is upper-lobe centriacinar/centrilobular). Failing to check an AAT level in early-onset emphysema is the miss.

10. Liver knight (alpha-1 liver disease)

WHAT YOU SEE

Armored knight at the alpha-1 station representing hepatic involvement.

WHAT IT ENCODES

Alpha-1 antitrypsin deficiency also causes LIVER disease — misfolded mutant Z protein accumulates in hepatocytes (PAS-positive, diastase-resistant globules) causing neonatal cholestasis, cirrhosis, and hepatocellular carcinoma. Lung disease comes from too little circulating AAT; liver disease from accumulation of the abnormal protein.

BOARD TRAP

Suspect alpha-1 when emphysema coexists with unexplained liver disease in a young patient. The liver damage is from protein ACCUMULATION (gain of toxicity), distinct from the lung's loss-of-function mechanism.

11. Cor pulmonale / clubbing think cancer

WHAT YOU SEE

COR PULMONALE panel: FEV1<25%, low PaO2; note 'clubbing + COPD → think lung cancer'.

WHAT IT ENCODES

Severe hypoxemic COPD (FEV1 <25%, chronic low PaO2) drives pulmonary hypertension and cor pulmonale (right heart failure). Digital CLUBBING is NOT a feature of COPD itself and should prompt a search for another cause.

BOARD TRAP

Clubbing in a COPD patient is a RED FLAG for lung cancer (or bronchiectasis/ILD) — do not attribute it to COPD; image the chest.

12. Roflumilast PDE4 inhibitor

WHAT YOU SEE

PDE4 panel for chronic bronchitis phenotype with FEV1<50% and exacerbations.

WHAT IT ENCODES

Roflumilast is an oral PDE4 inhibitor that reduces exacerbations in the CHRONIC BRONCHITIS phenotype with FEV1 < 50% and frequent exacerbations despite inhaled therapy. It raises intracellular cAMP, reducing airway inflammation.

BOARD TRAP

Roflumilast is reserved for the chronic-bronchitis/exacerbator phenotype, not all COPD. Key side effects: weight loss, nausea/diarrhea, and psychiatric effects (mood changes) — avoid in underweight or depressed patients.

13. Azithromycin triple therapy syringe

WHAT YOU SEE

Syringe panel: 'EOS >=100, triple therapy, still exacerbating' with macrolide.

WHAT IT ENCODES

Chronic low-dose azithromycin (immunomodulatory/anti-inflammatory plus antimicrobial effect) reduces exacerbation frequency in patients STILL exacerbating on maximal inhaled (triple) therapy, especially former smokers.

BOARD TRAP

Add-on chronic azithromycin is for REFRACTORY exacerbators, not routine maintenance. Watch for QT prolongation, hearing loss, and macrolide resistance; screen for/exclude active NTM infection first.

14. Lung volume reduction surgery (LVRS)

WHAT YOU SEE

Surgeon panel 'LVRS': upper-lobe emphysema, low exercise capacity = best candidates.

WHAT IT ENCODES

Lung volume reduction surgery removes the most destroyed (hyperinflated) tissue to improve elastic recoil and diaphragm mechanics. Best candidates have UPPER-lobe predominant emphysema AND low baseline exercise capacity — this subgroup gets a survival and functional benefit.

BOARD TRAP

LVRS helps upper-lobe disease with poor exercise tolerance. It is HARMFUL (increased mortality) in patients with FEV1 <20% plus either homogeneous emphysema or very low DLCO — these are contraindications, not candidates.